Newman-Penrose Formalism Calculations

Stephani et al., Equation (23.14a)

Metric

$$ds^2 = \left(-G\left(x\right)^{-2\,m+1} P\left(x\right)^n T\left(t\right)^{2\,m}\right) \mathrm{d}t \otimes \mathrm{d}t + \left(G\left(x\right)^{-2\,m+1} P\left(x\right)^n T\left(t\right)^{2\,m} H\left(x\right)^2\right) \mathrm{d}x \otimes \mathrm{d}x + \left(T\left(t\right)^{n+1} G\left(x\right) P\left(x\right)\right) \mathrm{d}y \otimes \mathrm{d}y + \left(\frac{T\left(t\right)^{-n+1} G\left(x\right)}{P\left(x\right)}\right) \mathrm{d}z \otimes \mathrm{d}z$$

Coordinates

$(x^\mu) = (t, x, y, z)$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2}\sqrt{2}\sqrt{G\left(x\right)^{-2\,m} P\left(x\right)^n G\left(x\right)} T\left(t\right)^m\right) dt + \left(\frac{1}{2}\sqrt{2}\sqrt{G\left(x\right)^{-2\,m} P\left(x\right)^n G\left(x\right)} T\left(t\right)^m H\left(x\right)\right) dx$$

$$n = \left(\frac{1}{2}\sqrt{2}\sqrt{G\left(x\right)^{-2\,m} P\left(x\right)^n G\left(x\right)} T\left(t\right)^m\right) dt + \left(-\frac{1}{2}\sqrt{2}\sqrt{G\left(x\right)^{-2\,m} P\left(x\right)^n G\left(x\right)} T\left(t\right)^m H\left(x\right)\right) dx$$

$$m = \left(\frac{1}{2}\sqrt{2} T\left(t\right)^{\frac{1}{2}\,n} \sqrt{G\left(x\right)}\sqrt{P\left(x\right)}\sqrt{T\left(t\right)}\right) dy + \left(\frac{i\sqrt{2} T\left(t\right)^{-\frac{1}{2}\,n} \sqrt{G\left(x\right)}\sqrt{T\left(t\right)}}{2\sqrt{P\left(x\right)}}\right) dz$$

$$\overline{m} = \left(\frac{1}{2}\sqrt{2} T\left(t\right)^{\frac{1}{2}\,n} \sqrt{G\left(x\right)}\sqrt{P\left(x\right)}\sqrt{T\left(t\right)}\right) dy + \left(-\frac{i\sqrt{2} T\left(t\right)^{-\frac{1}{2}\,n} \sqrt{G\left(x\right)}\sqrt{T\left(t\right)}}{2\sqrt{P\left(x\right)}}\right) dz$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}G\left(x\right)^{2\,m}\left|G\left(x\right)\right|^{-m}\left|T\left(t\right)\right|^{m}\sqrt{\left|G\left(x\right)\right|}}{2\,\sqrt{P\left(x\right)^{n}}T\left(t\right)^{2\,m}G\left(x\right)}\right)dt + \left(\frac{\sqrt{2}G\left(x\right)^{2\,m}\left|G\left(x\right)\right|^{-m}\left|T\left(t\right)\right|^{m}\sqrt{\left|G\left(x\right)\right|}}{2\,\sqrt{P\left(x\right)^{n}}T\left(t\right)^{2\,m}G\left(x\right)H\left(x\right)}\right)dx$$

$$n = \left(-\frac{\sqrt{2}G\left(x\right)^{2\,m}\left|G\left(x\right)\right|^{-m}\left|T\left(t\right)\right|^{m}\sqrt{\left|G\left(x\right)\right|}}{2\,\sqrt{P\left(x\right)^{n}}T\left(t\right)^{2\,m}G\left(x\right)}\right)dt + \left(-\frac{\sqrt{2}G\left(x\right)^{2\,m}\left|G\left(x\right)\right|^{-m}\left|T\left(t\right)\right|^{m}\sqrt{\left|G\left(x\right)\right|}}{2\,\sqrt{P\left(x\right)^{n}}T\left(t\right)^{2\,m}G\left(x\right)H\left(x\right)}\right)dx$$

$$m = \left(\frac{\sqrt{2}T\left(t\right)^{-\frac{1}{2}\,n}}{2\,\sqrt{G\left(x\right)}\sqrt{P\left(x\right)}\sqrt{T\left(t\right)}}\right)dy + \left(\frac{i\,\sqrt{2}T\left(t\right)^{\frac{1}{2}\,n}\sqrt{P\left(x\right)}}{2\,\sqrt{G\left(x\right)}\sqrt{T\left(t\right)}}\right)dz$$

$$\overline{m} = \left(\frac{\sqrt{2}T\left(t\right)^{-\frac{1}{2}\,n}}{2\,\sqrt{G\left(x\right)}\sqrt{P\left(x\right)}\sqrt{T\left(t\right)}}\right)dy + \left(-\frac{i\,\sqrt{2}T\left(t\right)^{\frac{1}{2}\,n}\sqrt{P\left(x\right)}}{2\,\sqrt{G\left(x\right)}\sqrt{T\left(t\right)}}\right)dz$$

Directional Derivatives

$$D = -\frac{\sqrt{2}G\left(x\right)^{2\,m}\left|G\left(x\right)\right|^{-m}\left|T\left(t\right)\right|^{m}\sqrt{\left|G\left(x\right)\right|}}{2\,\sqrt{P\left(x\right)^{n}}T\left(t\right)^{2\,m}G\left(x\right)}\,\partial_{t} + \frac{\sqrt{2}G\left(x\right)^{2\,m}\left|G\left(x\right)\right|^{-m}\left|T\left(t\right)\right|^{m}\sqrt{\left|G\left(x\right)\right|}}{2\,\sqrt{P\left(x\right)^{n}}T\left(t\right)^{2\,m}G\left(x\right)H\left(x\right)}\,\partial_{x}$$

$$\Delta = -\frac{\sqrt{2}G\left(x\right)^{2\,m}\left|G\left(x\right)\right|^{-m}\left|T\left(t\right)\right|^{m}\sqrt{\left|G\left(x\right)\right|}}{2\,\sqrt{P\left(x\right)^{n}}T\left(t\right)^{2\,m}G\left(x\right)}\,\partial_{t} - \frac{\sqrt{2}G\left(x\right)^{2\,m}\left|G\left(x\right)\right|^{-m}\left|T\left(t\right)\right|^{m}\sqrt{\left|G\left(x\right)\right|}}{2\,\sqrt{P\left(x\right)^{n}}T\left(t\right)^{2\,m}G\left(x\right)H\left(x\right)}\,\partial_{x}$$

$$\delta = \frac{\sqrt{2}T\left(t\right)^{-\frac{1}{2}\,n}}{2\,\sqrt{G\left(x\right)}\sqrt{P\left(x\right)}\sqrt{T\left(t\right)}}\,\partial_{y} + \frac{i\,\sqrt{2}T\left(t\right)^{\frac{1}{2}\,n}\sqrt{P\left(x\right)}}{2\,\sqrt{G\left(x\right)}\sqrt{T\left(t\right)}}\,\partial_{z}$$

$$\overline{\delta} = \frac{\sqrt{2}T\left(t\right)^{-\frac{1}{2}\,n}}{2\,\sqrt{G\left(x\right)}\sqrt{P\left(x\right)}\sqrt{T\left(t\right)}}\,\partial_{y} - \frac{i\,\sqrt{2}T\left(t\right)^{\frac{1}{2}\,n}\sqrt{P\left(x\right)}}{2\,\sqrt{G\left(x\right)}\sqrt{T\left(t\right)}}\,\partial_{z}$$

Spin Coefficients

$$\rho = \frac{\sqrt{2}G\left(x\right)^{2\,m}T\left(t\right)^{-2\,m}\left|G\left(x\right)\right|^{-m}\left|T\left(t\right)\right|^{m}\sqrt{\left|G\left(x\right)\right|}\frac{\partial}{\partial t}T\left(t\right)}{4\sqrt{P\left(x\right)^{n}}G\left(x\right)T\left(t\right)} - \frac{\sqrt{2}G\left(x\right)^{2\,m}T\left(t\right)^{-2\,m}\left|G\left(x\right)\right|^{-m}\left|T\left(t\right)\right|^{m}\sqrt{\left|G\left(x\right)\right|}\frac{\partial}{\partial x}G\left(x\right)}{4\sqrt{P\left(x\right)^{n}}G\left(x\right)^{2}H\left(x\right)}$$

$$\sigma = \frac{\sqrt{2}nG\left(x\right)^{2\,m}T\left(t\right)^{-2\,m}\left|G\left(x\right)\right|^{-m}\left|T\left(t\right)\right|^{m}\sqrt{\left|G\left(x\right)\right|}\frac{\partial}{\partial t}T\left(t\right)}{4\sqrt{P\left(x\right)^{n}}G\left(x\right)T\left(t\right)} - \frac{\sqrt{2}G\left(x\right)^{2\,m}T\left(t\right)^{-2\,m}\left|G\left(x\right)\right|^{-m}\left|T\left(t\right)\right|^{m}\sqrt{\left|G\left(x\right)\right|}\frac{\partial}{\partial x}P\left(x\right)}{4\sqrt{P\left(x\right)^{n}}G\left(x\right)H\left(x\right)P\left(x\right)}$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}G\left(x\right)^{2\,m}T\left(t\right)^{-2\,m}\left|G\left(x\right)\right|^{-m}\left|T\left(t\right)\right|^{m}\sqrt{\left|G\left(x\right)\right|}\frac{\partial}{\partial t}T\left(t\right)}{4\sqrt{P\left(x\right)^{n}}G\left(x\right)T\left(t\right)} - \frac{\sqrt{2}G\left(x\right)^{2\,m}T\left(t\right)^{-2\,m}\left|G\left(x\right)\right|^{-m}\left|T\left(t\right)\right|^{m}\sqrt{\left|G\left(x\right)\right|}\frac{\partial}{\partial x}G\left(x\right)}{4\sqrt{P\left(x\right)^{n}}G\left(x\right)^{2}H\left(x\right)}$$

$$\lambda = -\frac{\sqrt{2}nG\left(x\right)^{2\,m}T\left(t\right)^{-2\,m}\left|G\left(x\right)\right|^{-m}\left|T\left(t\right)\right|^{m}\sqrt{\left|G\left(x\right)\right|}\frac{\partial}{\partial t}T\left(t\right)}{4\sqrt{P\left(x\right)^{n}}G\left(x\right)T\left(t\right)} - \frac{\sqrt{2}G\left(x\right)^{2\,m}T\left(t\right)^{-2\,m}\left|G\left(x\right)\right|^{-m}\left|T\left(t\right)\right|^{m}\sqrt{\left|G\left(x\right)\right|}\frac{\partial}{\partial x}P\left(x\right)}{4\sqrt{P\left(x\right)^{n}}G\left(x\right)H\left(x\right)P\left(x\right)}$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$
\begin{aligned}
\epsilon = & -\frac{\sqrt{2} m G(x)^{2m} P(x)^{-n} \sqrt{P(x)^{n}} T(t)^{2m} T(t)^{-4m} |G(x)|^{2m} |G(x)|^{-3m} |T(t)|^{m} \sqrt{|G(x)|} \frac{\partial}{\partial t} T(t)}{4 G(x) T(t)} \\
& - \frac{\sqrt{2} m G(x)^{2m} P(x)^{-n} \sqrt{P(x)^{n}} T(t)^{2m} T(t)^{-4m} |G(x)|^{2m} |G(x)|^{-3m} |T(t)|^{m} \sqrt{|G(x)|} \frac{\partial}{\partial x} G(x)}{4 G(x)^{2} H(x)} \\
& + \frac{\sqrt{2} n G(x)^{2m} P(x)^{-n} \sqrt{P(x)^{n}} T(t)^{2m} T(t)^{-4m} |G(x)|^{2m} |G(x)|^{-3m} |T(t)|^{m} \sqrt{|G(x)|} \frac{\partial}{\partial x} P(x)}{8 G(x) H(x) P(x)} \\
& + \frac{\sqrt{2} G(x)^{2m} P(x)^{-n} \sqrt{P(x)^{n}} T(t)^{2m} T(t)^{-4m} |G(x)|^{2m} |G(x)|^{-3m} |T(t)|^{m} \sqrt{|G(x)|} \frac{\partial}{\partial x} G(x)}{8 G(x)^{2} H(x)} \\
& + \frac{\sqrt{2} G(x)^{2m} P(x)^{-n} \sqrt{P(x)^{n}} T(t)^{2m} T(t)^{-4m} |G(x)|^{2m} |G(x)|^{-3m} |T(t)|^{m} \sqrt{|G(x)|} \frac{\partial}{\partial x} H(x)}{8 G(x) H(x)^{2}} \\
& - \frac{\sqrt{2} G(x)^{4m} P(x)^{-n} \sqrt{P(x)^{n}} T(t)^{-4m} |G(x)|^{-3m} |T(t)|^{3m} |G(x)|^{\frac{3}{2}} \frac{\partial}{\partial x} H(x)}{8 G(x)^{2} H(x)^{2}}
\end{aligned}
$$

$$
\begin{aligned}
\gamma = & \frac{\sqrt{2} m G(x)^{2m} P(x)^{-n} \sqrt{P(x)^{n}} T(t)^{2m} T(t)^{-4m} |G(x)|^{2m} |G(x)|^{-3m} |T(t)|^{m} \sqrt{|G(x)|} \frac{\partial}{\partial t} T(t)}{4 G(x) T(t)} \\
& - \frac{\sqrt{2} m G(x)^{2m} P(x)^{-n} \sqrt{P(x)^{n}} T(t)^{2m} T(t)^{-4m} |G(x)|^{2m} |G(x)|^{-3m} |T(t)|^{m} \sqrt{|G(x)|} \frac{\partial}{\partial x} G(x)}{4 G(x)^{2} H(x)} \\
& + \frac{\sqrt{2} n G(x)^{2m} P(x)^{-n} \sqrt{P(x)^{n}} T(t)^{2m} T(t)^{-4m} |G(x)|^{2m} |G(x)|^{-3m} |T(t)|^{m} \sqrt{|G(x)|} \frac{\partial}{\partial x} P(x)}{8 G(x) H(x) P(x)} \\
& + \frac{\sqrt{2} G(x)^{2m} P(x)^{-n} \sqrt{P(x)^{n}} T(t)^{2m} T(t)^{-4m} |G(x)|^{2m} |G(x)|^{-3m} |T(t)|^{m} \sqrt{|G(x)|} \frac{\partial}{\partial x} G(x)}{8 G(x)^{2} H(x)} \\
& + \frac{\sqrt{2} G(x)^{2m} P(x)^{-n} \sqrt{P(x)^{n}} T(t)^{2m} T(t)^{-4m} |G(x)|^{2m} |G(x)|^{-3m} |T(t)|^{m} \sqrt{|G(x)|} \frac{\partial}{\partial x} H(x)}{8 G(x) H(x)^{2}} \\
& - \frac{\sqrt{2} G(x)^{4m} P(x)^{-n} \sqrt{P(x)^{n}} T(t)^{-4m} |G(x)|^{-3m} |T(t)|^{3m} |G(x)|^{\frac{3}{2}} \frac{\partial}{\partial x} H(x)}{8 G(x)^{2} H(x)^{2}}
\end{aligned}
$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{n^2 G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial t} T(t)^2}{8 |G(x)|^{2m} G(x)^2 T(t)^2} + \frac{m G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial t} T(t)^2}{2 |G(x)|^{2m} G(x)^2 T(t)^2}$$
$$+ \frac{G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial t} T(t)^2}{8 |G(x)|^{2m} G(x)^2 T(t)^2} - \frac{G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial^2}{(\partial t)^2} T(t)}{4 |G(x)|^{2m} G(x)^2 T(t)}$$
$$- \frac{m G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} G(x)^2}{2 |G(x)|^{2m} G(x)^4 H(x)^2} + \frac{n G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial x} P(x)}{4 |G(x)|^{2m} G(x)^3 H(x)^2 P(x)}$$
$$+ \frac{3 G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} G(x)^2}{8 |G(x)|^{2m} G(x)^4 H(x)^2} - \frac{G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial^2}{(\partial x)^2} G(x)}{4 |G(x)|^{2m} G(x)^3 H(x)^2}$$
$$+ \frac{G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial x} H(x)}{4 |G(x)|^{2m} G(x)^3 H(x)^3} - \frac{G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} P(x)^2}{8 |G(x)|^{2m} G(x)^2 H(x)^2 P(x)^2}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = \frac{n G(x)^{2m} P(x)^{-n} T(t)^{-2m} \frac{\partial^2}{(\partial t)^2} T(t)}{4 G(x) T(t)} - \frac{G(x)^{2m} P(x)^{-n} T(t)^{-2m} \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial x} P(x)}{4 G(x)^2 H(x)^2 P(x)}$$
$$+ \frac{G(x)^{2m} P(x)^{-n} T(t)^{-2m} \frac{\partial}{\partial x} H(x) \frac{\partial}{\partial x} P(x)}{4 G(x) H(x)^3 P(x)} + \frac{G(x)^{2m} P(x)^{-n} T(t)^{-2m} \frac{\partial}{\partial x} P(x)^2}{4 G(x) H(x)^2 P(x)^2} - \frac{G(x)^{2m} P(x)^{-n} T(t)^{-2m} \frac{\partial^2}{(\partial x)^2} P(x)}{4 G(x) H(x)^2 P(x)}$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{n^2 G(x)^{2m} P(x)^{-n} T(t)^{2m} T(t)^{-4m} \frac{\partial}{\partial t} T(t)^2}{32\, G(x)\, T(t)^2} + \frac{m G(x)^{2m} P(x)^{-n} T(t)^{2m} T(t)^{-4m} \frac{\partial}{\partial t} T(t)^2}{8\, G(x)\, T(t)^2} - \frac{n^2 G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial t} T(t)^2}{32\, |G(x)|^{2m} G(x)^2\, T(t)^2}$$
$$- \frac{m G(x)^{2m} P(x)^{-n} T(t)^{2m} T(t)^{-4m} \frac{\partial^2}{(\partial t)^2} T(t)}{8\, G(x)\, T(t)} + \frac{G(x)^{2m} P(x)^{-n} T(t)^{2m} T(t)^{-4m} \frac{\partial}{\partial t} T(t)^2}{32\, G(x)\, T(t)^2} + \frac{m G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial t} T(t)^2}{8\, |G(x)|^{2m} G(x)^2\, T(t)^2}$$
$$- \frac{m G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial^2}{(\partial t)^2} T(t)}{8\, |G(x)|^{2m} G(x)^2\, T(t)} + \frac{m G(x)^{2m} P(x)^{-n} T(t)^{2m} T(t)^{-4m} \frac{\partial}{\partial x} G(x)^2}{8\, G(x)^3\, H(x)^2} - \frac{m G(x)^{2m} P(x)^{-n} T(t)^{2m} T(t)^{-4m} \frac{\partial^2}{(\partial x)^2} G(x)}{8\, G(x)^2\, H(x)^2}$$
$$+ \frac{m G(x)^{2m} P(x)^{-n} T(t)^{2m} T(t)^{-4m} \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial x} H(x)}{8\, G(x)^2\, H(x)^3} - \frac{n G(x)^{2m} P(x)^{-n} T(t)^{2m} T(t)^{-4m} \frac{\partial}{\partial x} H(x) \frac{\partial}{\partial x} P(x)}{16\, G(x)\, H(x)^3\, P(x)} - \frac{n G(x)^{2m} P(x)^{-n} T(t)^{2m} T(t)^{-4m} \frac{\partial}{\partial x} P(x)^2}{16\, G(x)\, H(x)^2\, P(x)^2}$$
$$+ \frac{n G(x)^{2m} P(x)^{-n} T(t)^{2m} T(t)^{-4m} \frac{\partial^2}{(\partial x)^2} P(x)}{16\, G(x)\, H(x)^2\, P(x)} + \frac{G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial t} T(t)^2}{32\, |G(x)|^{2m} G(x)^2\, T(t)^2} - \frac{3\, G(x)^{2m} P(x)^{-n} T(t)^{2m} T(t)^{-4m} \frac{\partial}{\partial x} G(x)^2}{32\, G(x)^3\, H(x)^2}$$
$$+ \frac{m G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} G(x)^2}{8\, |G(x)|^{2m} G(x)^4\, H(x)^2} + \frac{G(x)^{2m} P(x)^{-n} T(t)^{2m} T(t)^{-4m} \frac{\partial^2}{(\partial x)^2} G(x)}{16\, G(x)^2\, H(x)^2} - \frac{m G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial^2}{(\partial x)^2} G(x)}{8\, |G(x)|^{2m} G(x)^3\, H(x)^2}$$
$$- \frac{G(x)^{2m} P(x)^{-n} T(t)^{2m} T(t)^{-4m} \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial x} H(x)}{16\, G(x)^2\, H(x)^3} + \frac{m G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial x} H(x)}{8\, |G(x)|^{2m} G(x)^3\, H(x)^3} - \dots$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = \frac{n G(x)^{2m} P(x)^{-n} T(t)^{-2m} \frac{\partial^2}{(\partial t)^2} T(t)}{4\, G(x)\, T(t)} - \frac{G(x)^{2m} P(x)^{-n} T(t)^{-2m} \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial x} P(x)}{4\, G(x)^2\, H(x)^2\, P(x)}$$
$$+ \frac{G(x)^{2m} P(x)^{-n} T(t)^{-2m} \frac{\partial}{\partial x} H(x) \frac{\partial}{\partial x} P(x)}{4\, G(x)\, H(x)^3\, P(x)} + \frac{G(x)^{2m} P(x)^{-n} T(t)^{-2m} \frac{\partial}{\partial x} P(x)^2}{4\, G(x)\, H(x)^2\, P(x)^2} - \frac{G(x)^{2m} P(x)^{-n} T(t)^{-2m} \frac{\partial^2}{(\partial x)^2} P(x)}{4\, G(x)\, H(x)^2\, P(x)}$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = -\frac{n^2 G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial t} T(t)^2}{8 |G(x)|^{2m} G(x)^2 T(t)^2} + \frac{m G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial t} T(t)^2}{2 |G(x)|^{2m} G(x)^2 T(t)^2}$$
$$+ \frac{G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial t} T(t)^2}{8 |G(x)|^{2m} G(x)^2 T(t)^2} - \frac{G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial^2}{(\partial t)^2} T(t)}{4 |G(x)|^{2m} G(x)^2 T(t)}$$
$$- \frac{m G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} G(x)^2}{2 |G(x)|^{2m} G(x)^4 H(x)^2} + \frac{n G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial x} P(x)}{4 |G(x)|^{2m} G(x)^3 H(x)^2 P(x)}$$
$$+ \frac{3 G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} G(x)^2}{8 |G(x)|^{2m} G(x)^4 H(x)^2} - \frac{G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial^2}{(\partial x)^2} G(x)}{4 |G(x)|^{2m} G(x)^3 H(x)^2}$$
$$+ \frac{G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial x} H(x)}{4 |G(x)|^{2m} G(x)^3 H(x)^3} - \frac{G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} P(x)^2}{8 |G(x)|^{2m} G(x)^2 H(x)^2 P(x)^2}$$

$$\Lambda = \frac{n^2 G(x)^{2m} P(x)^{-n} T(t)^{-2m} \frac{\partial}{\partial t} T(t)^2}{48 G(x) T(t)^2} - \frac{m G(x)^{2m} P(x)^{-n} T(t)^{-2m} \frac{\partial}{\partial t} T(t)^2}{12 G(x) T(t)^2} + \frac{m G(x)^{2m} P(x)^{-n} T(t)^{-2m} \frac{\partial^2}{(\partial t)^2} T(t)}{12 G(x) T(t)} - \frac{G(x)^{2m} P(x)^{-n} T(t)^{-2m} \frac{\partial}{\partial t} T(t)^2}{48 G(x) T(t)^2}$$
$$+ \frac{G(x)^{2m} P(x)^{-n} T(t)^{-2m} \frac{\partial^2}{(\partial t)^2} T(t)}{12 G(x) T(t)} - \frac{m G(x)^{2m} P(x)^{-n} T(t)^{-2m} \frac{\partial}{\partial x} G(x)^2}{12 G(x)^3 H(x)^2} + \frac{m G(x)^{2m} P(x)^{-n} T(t)^{-2m} \frac{\partial^2}{(\partial x)^2} G(x)}{12 G(x)^2 H(x)^2} - \frac{m G(x)^{2m} P(x)^{-n} T(t)^{-2m} \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial x} H(x)}{12 G(x)^2 H(x)^3}$$
$$+ \frac{n G(x)^{2m} P(x)^{-n} T(t)^{-2m} \frac{\partial}{\partial x} H(x) \frac{\partial}{\partial x} P(x)}{24 G(x) H(x)^3 P(x)} + \frac{n G(x)^{2m} P(x)^{-n} T(t)^{-2m} \frac{\partial}{\partial x} P(x)^2}{24 G(x) H(x)^2 P(x)^2} - \frac{n G(x)^{2m} P(x)^{-n} T(t)^{-2m} \frac{\partial^2}{(\partial x)^2} P(x)}{24 G(x) H(x)^2 P(x)}$$
$$+ \frac{G(x)^{2m} P(x)^{-n} T(t)^{-2m} \frac{\partial}{\partial x} G(x)^2}{16 G(x)^3 H(x)^2} - \frac{G(x)^{2m} P(x)^{-n} T(t)^{-2m} \frac{\partial^2}{(\partial x)^2} G(x)}{8 G(x)^2 H(x)^2} + \frac{G(x)^{2m} P(x)^{-n} T(t)^{-2m} \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial x} H(x)}{8 G(x)^2 H(x)^3} - \frac{G(x)^{2m} P(x)^{-n} T(t)^{-2m} \frac{\partial}{\partial x} P(x)^2}{48 G(x) H(x)^2 P(x)^2}$$

Weyl Tensor Components

$$\Psi_0 = \frac{mnG(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial t} T(t)^2}{2|G(x)|^{2m} G(x)^2 T(t)^2} + \frac{mnG(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial t} T(t)}{2|G(x)|^{2m} G(x)^3 H(x) T(t)}$$
$$- \frac{n^2 G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} P(x) \frac{\partial}{\partial t} T(t)}{4|G(x)|^{2m} G(x)^2 H(x) P(x) T(t)} - \frac{nG(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial^2}{(\partial t)^2} T(t)}{4|G(x)|^{2m} G(x)^2 T(t)}$$
$$- \frac{mG(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} P(x) \frac{\partial}{\partial t} T(t)}{2|G(x)|^{2m} G(x)^2 H(x) P(x) T(t)} - \frac{mG(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial x} P(x)}{2|G(x)|^{2m} G(x)^3 H(x)^2 P(x)}$$
$$+ \frac{nG(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} P(x)^2}{4|G(x)|^{2m} G(x)^2 H(x)^2 P(x)^2} + \frac{G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} P(x) \frac{\partial}{\partial t} T(t)}{4|G(x)|^{2m} G(x)^2 H(x) P(x) T(t)}$$
$$+ \frac{G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} H(x) \frac{\partial}{\partial x} P(x)}{4|G(x)|^{2m} G(x)^2 H(x)^3 P(x)}$$
$$+ \frac{G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} P(x)^2}{4|G(x)|^{2m} G(x)^2 H(x)^2 P(x)^2} - \frac{G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial^2}{(\partial x)^2} P(x)}{4|G(x)|^{2m} G(x)^2 H(x)^2 P(x)}$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{n^2 G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial t} T(t)^2}{12|G(x)|^{2m} G(x)^2 T(t)^2} + \frac{mG(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial t} T(t)^2}{6|G(x)|^{2m} G(x)^2 T(t)^2} - \frac{mG(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial^2}{(\partial t)^2} T(t)}{6|G(x)|^{2m} G(x)^2 T(t)}$$
$$- \frac{G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial t} T(t)^2}{12|G(x)|^{2m} G(x)^2 T(t)^2} + \frac{G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial^2}{(\partial t)^2} T(t)}{12|G(x)|^{2m} G(x)^2 T(t)} + \frac{mG(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} G(x)^2}{6|G(x)|^{2m} G(x)^4 H(x)^2}$$
$$- \frac{mG(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial^2}{(\partial x)^2} G(x)}{6|G(x)|^{2m} G(x)^3 H(x)^2} + \frac{mG(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial x} H(x)}{6|G(x)|^{2m} G(x)^3 H(x)^3}$$
$$- \frac{nG(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} H(x) \frac{\partial}{\partial x} P(x)}{12|G(x)|^{2m} G(x)^2 H(x)^3 P(x)} - \frac{nG(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} P(x)^2}{12|G(x)|^{2m} G(x)^2 H(x)^2 P(x)^2}$$
$$+ \frac{nG(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial^2}{(\partial x)^2} P(x)}{12|G(x)|^{2m} G(x)^2 H(x)^2 P(x)} - \frac{G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} P(x)^2}{12|G(x)|^{2m} G(x)^2 H(x)^2 P(x)^2}$$

$$\Psi_3 = 0$$

$$\Psi_4 = \frac{mnG(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial t} T(t)^2}{2|G(x)|^{2m} G(x)^2 T(t)^2} - \frac{mnG(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial t} T(t)}{2|G(x)|^{2m} G(x)^3 H(x) T(t)}$$
$$+ \frac{n^2 G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} P(x) \frac{\partial}{\partial t} T(t)}{4|G(x)|^{2m} G(x)^2 H(x) P(x) T(t)} - \frac{nG(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial^2}{(\partial t)^2} T(t)}{4|G(x)|^{2m} G(x)^2 T(t)}$$
$$+ \frac{mG(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} P(x) \frac{\partial}{\partial t} T(t)}{2|G(x)|^{2m} G(x)^2 H(x) P(x) T(t)} - \frac{mG(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial x} P(x)}{2|G(x)|^{2m} G(x)^3 H(x)^2 P(x)}$$
$$+ \frac{nG(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} P(x)^2}{4|G(x)|^{2m} G(x)^2 H(x)^2 P(x)^2} - \frac{G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} P(x) \frac{\partial}{\partial t} T(t)}{4|G(x)|^{2m} G(x)^2 H(x) P(x) T(t)}$$
$$+ \frac{G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} H(x) \frac{\partial}{\partial x} P(x)}{4|G(x)|^{2m} G(x)^2 H(x)^3 P(x)}$$
$$+ \frac{G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} P(x)^2}{4|G(x)|^{2m} G(x)^2 H(x)^2 P(x)^2} - \frac{G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial^2}{(\partial x)^2} P(x)}{4|G(x)|^{2m} G(x)^2 H(x)^2 P(x)}$$

Newman-Penrose Equations

$$NP1 = \frac{mG(x)^{4m} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{\frac{\partial}{\partial t} T(t)} \frac{\partial}{\partial t} T(t)}{8P(x)^n T(t)^2 |G(x)|} - \frac{mG(x)^{4m} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \frac{\partial}{\partial t} T(t)^2}{4P(x)^n T(t)^2 |G(x)|}$$
$$+ \frac{mG(x)^{4m} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{T(t)} \frac{\partial}{\partial t} T(t)^2}{8P(x)^n T(t)^3 |G(x)|} - \frac{mG(x)^{4m} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{\frac{\partial}{\partial t} T(t)} \frac{\partial}{\partial x} G(x)}{8P(x)^n G(x) H(x) T(t) |G(x)|}$$
$$+ \frac{mG(x)^{4m} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{\frac{\partial}{\partial x} G(x)} \frac{\partial}{\partial t} T(t)}{8P(x)^n G(x) H(x) T(t) |G(x)|} + \frac{mG(x)^{4m} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{G(x)} \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial t} T(t)}{8P(x)^n G(x)^2 H(x) T(t) |G(x)|}$$
$$- \frac{mG(x)^{4m} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{T(t)} \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial t} T(t)}{8P(x)^n G(x) H(x) T(t)^2 |G(x)|} - \frac{mG(x)^{4m} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{\frac{\partial}{\partial x} G(x)} \frac{\partial}{\partial x} G(x)}{8P(x)^n G(x)^2 H(x)^2 |G(x)|}$$
$$+ \frac{mG(x)^{4m} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \frac{\partial}{\partial x} G(x)^2}{4P(x)^n G(x)^2 H(x)^2 |G(x)|} - \frac{mG(x)^{4m} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{G(x)} \frac{\partial}{\partial x} G(x)^2}{8P(x)^n G(x)^3 H(x)^2 |G(x)|}$$
$$- \frac{G(x)^{4m} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{\frac{\partial}{\partial x} G(x)} \frac{\partial}{\partial t} T(t)}{16P(x)^n G(x) H(x) T(t) |G(x)|} + \frac{G(x)^{4m} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial t} T(t)}{8P(x)^n G(x) H(x) T(t) |G(x)|} - \dots$$

$$
\begin{aligned}
NP2 = {} & \frac{mnG(x)^{4m} P(x)^{-n} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{\frac{\partial}{\partial t} T(t)} \frac{\partial}{\partial t} T(t)}{8 T(t)^2 |G(x)|} - \frac{mnG(x)^{4m} P(x)^{-n} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \frac{\partial}{\partial t} T(t)^2}{4 T(t)^2 |G(x)|} \\
& + \frac{mnG(x)^{4m} P(x)^{-n} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{T(t)} \frac{\partial}{\partial t} T(t)^2}{8 T(t)^3 |G(x)|} \\
& + \frac{mnG(x)^{4m} P(x)^{-n} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{\frac{\partial}{\partial x} G(x)} \frac{\partial}{\partial t} T(t)}{8 G(x) H(x) T(t) |G(x)|} \\
& - \frac{mnG(x)^{4m} P(x)^{-n} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial t} T(t)}{4 G(x) H(x) T(t) |G(x)|} \\
& + \frac{mnG(x)^{4m} P(x)^{-n} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{G(x)} \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial t} T(t)}{8 G(x)^2 H(x) T(t) |G(x)|} \\
& - \frac{mG(x)^{4m} P(x)^{-n} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{\frac{\partial}{\partial t} T(t)} \frac{\partial}{\partial x} P(x)}{8 H(x) P(x) T(t) |G(x)|} \\
& - \frac{nG(x)^{4m} P(x)^{-n} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{\frac{\partial}{\partial x} G(x)} \frac{\partial}{\partial t} T(t)}{16 G(x) H(x) T(t) |G(x)|} \\
& + \frac{nG(x)^{4m} P(x)^{-n} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial t} T(t)}{8 G(x) H(x) T(t) |G(x)|} \\
& - \frac{nG(x)^{4m} P(x)^{-n} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{G(x)} \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial t} T(t)}{16 G(x)^2 H(x) T(t) |G(x)|} \\
& + \frac{nG(x)^{4m} P(x)^{-n} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \frac{\partial}{\partial x} H(x) \frac{\partial}{\partial t} T(t)}{8 H(x)^2 T(t) |G(x)|} \\
& + \frac{mG(x)^{4m} P(x)^{-n} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \frac{\partial}{\partial x} P(x) \frac{\partial}{\partial t} T(t)}{4 H(x) P(x) T(t) |G(x)|} - \cdots
\end{aligned}
$$

$$
\begin{aligned}
NP6 = {} & \frac{m^2 G(x)^{4m} P(x)^{-n} T(t)^{4m} T(t)^{-8m} |G(x)|^{4m} |G(x)|^{-6m} |T(t)|^{2m} \overline{\frac{\partial}{\partial t} T(t)} \frac{\partial}{\partial t} T(t)}{4 T(t)^2 |G(x)|} - \frac{m^2 G(x)^{4m} P(x)^{-n} T(t)^{4m} T(t)^{-8m} |G(x)|^{4m} |G(x)|^{-6m} |T(t)|^{2m} \frac{\partial}{\partial t} T(t)^2}{2 T(t)^2 |G(x)|} \\
& + \frac{5 n^2 G(x)^{4m} P(x)^{-n} T(t)^{4m} T(t)^{-8m} |G(x)|^{4m} |G(x)|^{-6m} |T(t)|^{2m} \frac{\partial}{\partial t} T(t)^2}{96 T(t)^2 |G(x)|} + \frac{m^2 G(x)^{4m} P(x)^{-n} T(t)^{4m} T(t)^{-8m} |G(x)|^{4m} |G(x)|^{-6m} |T(t)|^{2m} \overline{T(t)} \frac{\partial}{\partial t} T(t)^2}{4 T(t)^3 |G(x)|} \\
& - \frac{5 n^2 G(x)^{2m} P(x)^{-n} T(t)^{6m} T(t)^{-8m} |G(x)|^{6m} |G(x)|^{-6m} \frac{\partial}{\partial t} T(t)^2}{96 G(x) T(t)^2} - \frac{5 m G(x)^{4m} P(x)^{-n} T(t)^{4m} T(t)^{-8m} |G(x)|^{4m} |G(x)|^{-6m} |T(t)|^{2m} \frac{\partial}{\partial t} T(t)^2}{24 T(t)^2 |G(x)|} \\
& + \frac{5 m G(x)^{4m} P(x)^{-n} T(t)^{4m} T(t)^{-8m} |G(x)|^{4m} |G(x)|^{-6m} |T(t)|^{2m} \frac{\partial^2}{(\partial t)^2} T(t)}{24 T(t) |G(x)|} - \frac{m^2 G(x)^{4m} P(x)^{-n} T(t)^{4m} T(t)^{-8m} |G(x)|^{4m} |G(x)|^{-6m} |T(t)|^{2m} \overline{\frac{\partial}{\partial x} G(x)} \frac{\partial}{\partial x} G(x)}{4 G(x)^2 H(x)^2 |G(x)|} \\
& + \frac{m^2 G(x)^{4m} P(x)^{-n} T(t)^{4m} T(t)^{-8m} |G(x)|^{4m} |G(x)|^{-6m} |T(t)|^{2m} \frac{\partial}{\partial x} G(x)^2}{2 G(x)^2 H(x)^2 |G(x)|} - \frac{m^2 G(x)^{4m} P(x)^{-n} T(t)^{4m} T(t)^{-8m} |G(x)|^{4m} |G(x)|^{-6m} |T(t)|^{2m} \overline{G(x)} \frac{\partial}{\partial x} G(x)^2}{4 G(x)^3 H(x)^2 |G(x)|} \\
& + \frac{m n G(x)^{4m} P(x)^{-n} T(t)^{4m} T(t)^{-8m} |G(x)|^{4m} |G(x)|^{-6m} |T(t)|^{2m} \overline{\frac{\partial}{\partial x} G(x)} \frac{\partial}{\partial x} P(x)}{8 G(x) H(x)^2 P(x) |G(x)|} \\
& - \frac{m n G(x)^{4m} P(x)^{-n} T(t)^{4m} T(t)^{-8m} |G(x)|^{4m} |G(x)|^{-6m} |T(t)|^{2m} \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial x} P(x)}{4 G(x) H(x)^2 P(x) |G(x)|} \\
& + \frac{m n G(x)^{4m} P(x)^{-n} T(t)^{4m} T(t)^{-8m} |G(x)|^{4m} |G(x)|^{-6m} |T(t)|^{2m} \overline{G(x)} \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial x} P(x)}{8 G(x)^2 H(x)^2 P(x) |G(x)|} + \dots
\end{aligned}
$$

$$
\begin{aligned}
NP7 = & -\frac{mnG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\overline{\frac{\partial}{\partial t}T(t)}\frac{\partial}{\partial t}T(t)}{8T(t)^{2}|G(x)|} \\
& + \frac{mnG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\frac{\partial}{\partial t}T(t)^{2}}{4T(t)^{2}|G(x)|} - \frac{mnG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\overline{T(t)}\frac{\partial}{\partial t}T(t)^{2}}{8T(t)^{3}|G(x)|} \\
& - \frac{mnG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\overline{\frac{\partial}{\partial x}G(x)}\frac{\partial}{\partial t}T(t)}{8G(x)H(x)T(t)|G(x)|} \\
& + \frac{mnG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\frac{\partial}{\partial x}G(x)\frac{\partial}{\partial t}T(t)}{4G(x)H(x)T(t)|G(x)|} \\
& - \frac{mnG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\overline{G(x)}\frac{\partial}{\partial x}G(x)\frac{\partial}{\partial t}T(t)}{8G(x)^{2}H(x)T(t)|G(x)|} \\
& - \frac{nG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\frac{\partial^{2}}{(\partial t)^{2}}T(t)}{4T(t)|G(x)|} - \frac{mG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\overline{\frac{\partial}{\partial t}T(t)}\frac{\partial}{\partial x}P(x)}{8H(x)P(x)T(t)|G(x)|} \\
& + \frac{nG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\overline{\frac{\partial}{\partial x}G(x)}\frac{\partial}{\partial t}T(t)}{16G(x)H(x)T(t)|G(x)|} \\
& - \frac{nG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\frac{\partial}{\partial x}G(x)\frac{\partial}{\partial t}T(t)}{8G(x)H(x)T(t)|G(x)|} \\
& + \frac{nG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\overline{G(x)}\frac{\partial}{\partial x}G(x)\frac{\partial}{\partial t}T(t)}{16G(x)^{2}H(x)T(t)|G(x)|} \\
& + \frac{nG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\frac{\partial}{\partial x}H(x)\frac{\partial}{\partial t}T(t)}{8H(x)^{2}T(t)|G(x)|} + \ldots
\end{aligned}
$$

$$
\begin{aligned}
NP8 = & -\frac{n^2 G(x)^{4m} P(x)^{-n} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \frac{\partial}{\partial t} T(t)^2}{24\, T(t)^2 |G(x)|} - \frac{m G(x)^{4m} P(x)^{-n} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{\frac{\partial}{\partial t} T(t)} \frac{\partial}{\partial t} T(t)}{8\, T(t)^2 |G(x)|} \\
& + \frac{n^2 G(x)^{2m} P(x)^{-n} T(t)^{4m} T(t)^{-6m} |G(x)|^{4m} |G(x)|^{-4m} \frac{\partial}{\partial t} T(t)^2}{24\, G(x) T(t)^2} + \frac{5\, m G(x)^{4m} P(x)^{-n} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \frac{\partial}{\partial t} T(t)^2}{12\, T(t)^2 |G(x)|} \\
& - \frac{m G(x)^{4m} P(x)^{-n} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{T(t)} \frac{\partial}{\partial t} T(t)^2}{8\, T(t)^3 |G(x)|} - \frac{m G(x)^{4m} P(x)^{-n} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \frac{\partial^2}{(\partial t)^2} T(t)}{6\, T(t) |G(x)|} \\
& - \frac{m G(x)^{4m} P(x)^{-n} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{\frac{\partial}{\partial t} T(t)} \frac{\partial}{\partial x} G(x)}{8\, G(x) H(x) T(t) |G(x)|} \\
& - \frac{m G(x)^{4m} P(x)^{-n} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{\frac{\partial}{\partial x} G(x)} \frac{\partial}{\partial t} T(t)}{8\, G(x) H(x) T(t) |G(x)|} \\
& + \frac{m G(x)^{4m} P(x)^{-n} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial t} T(t)}{2\, G(x) H(x) T(t) |G(x)|} \\
& - \frac{m G(x)^{4m} P(x)^{-n} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{G(x)} \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial t} T(t)}{8\, G(x)^2 H(x) T(t) |G(x)|} \\
& - \frac{m G(x)^{4m} P(x)^{-n} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{T(t)} \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial t} T(t)}{8\, G(x) H(x) T(t)^2 |G(x)|} \\
& - \frac{m G(x)^{2m} P(x)^{-n} T(t)^{4m} T(t)^{-6m} |G(x)|^{4m} |G(x)|^{-4m} \frac{\partial}{\partial t} T(t)^2}{6\, G(x) T(t)^2} + \frac{G(x)^{4m} P(x)^{-n} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \frac{\partial}{\partial t} T(t)^2}{24\, T(t)^2 |G(x)|} + \dots
\end{aligned}
$$

$$
\begin{aligned}
NP10 = & -\frac{mnG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\overline{\frac{\partial}{\partial t}T(t)}\frac{\partial}{\partial t}T(t)}{8T(t)^{2}|G(x)|} \\
& + \frac{mnG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\frac{\partial}{\partial t}T(t)^{2}}{4T(t)^{2}|G(x)|} - \frac{mnG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\overline{T(t)}\frac{\partial}{\partial t}T(t)^{2}}{8T(t)^{3}|G(x)|} \\
& + \frac{mnG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\overline{\frac{\partial}{\partial x}G(x)}\frac{\partial}{\partial t}T(t)}{8G(x)H(x)T(t)|G(x)|} \\
& - \frac{mnG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\frac{\partial}{\partial x}G(x)\frac{\partial}{\partial t}T(t)}{4G(x)H(x)T(t)|G(x)|} \\
& + \frac{mnG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\overline{G(x)}\frac{\partial}{\partial x}G(x)\frac{\partial}{\partial t}T(t)}{8G(x)^{2}H(x)T(t)|G(x)|} \\
& - \frac{mG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\overline{\frac{\partial}{\partial t}T(t)}\frac{\partial}{\partial x}P(x)}{8H(x)P(x)T(t)|G(x)|} \\
& - \frac{nG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\overline{\frac{\partial}{\partial x}G(x)}\frac{\partial}{\partial t}T(t)}{16G(x)H(x)T(t)|G(x)|} \\
& + \frac{nG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\frac{\partial}{\partial x}G(x)\frac{\partial}{\partial t}T(t)}{8G(x)H(x)T(t)|G(x)|} \\
& - \frac{nG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\overline{G(x)}\frac{\partial}{\partial x}G(x)\frac{\partial}{\partial t}T(t)}{16G(x)^{2}H(x)T(t)|G(x)|} \\
& + \frac{nG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\frac{\partial}{\partial x}H(x)\frac{\partial}{\partial t}T(t)}{8H(x)^{2}T(t)|G(x)|} \\
& + \frac{mG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\frac{\partial}{\partial x}P(x)\frac{\partial}{\partial t}T(t)}{4H(x)P(x)T(t)|G(x)|} - \cdots
\end{aligned}
$$

$$
\begin{aligned}
NP12 = & -\frac{n^2 G(x)^{2m} P(x)^{-n} T(t)^{2m} T(t)^{-4m} \frac{\partial}{\partial t} T(t)^2}{96\, G(x)\, T(t)^2} + \frac{m G(x)^{2m} P(x)^{-n} T(t)^{2m} T(t)^{-4m} \frac{\partial}{\partial t} T(t)^2}{24\, G(x)\, T(t)^2} + \frac{n^2 G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial t} T(t)^2}{96\, |G(x)|^{2m} G(x)^2\, T(t)^2} \\
& - \frac{m G(x)^{2m} P(x)^{-n} T(t)^{2m} T(t)^{-4m} \frac{\partial^2}{(\partial t)^2} T(t)}{24\, G(x)\, T(t)} + \frac{G(x)^{2m} P(x)^{-n} T(t)^{2m} T(t)^{-4m} \frac{\partial}{\partial t} T(t)^2}{96\, G(x)\, T(t)^2} - \frac{m G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial t} T(t)^2}{24\, |G(x)|^{2m} G(x)^2\, T(t)^2} \\
& + \frac{G(x)^{2m} P(x)^{-n} T(t)^{2m} T(t)^{-4m} \frac{\partial^2}{(\partial t)^2} T(t)}{12\, G(x)\, T(t)} + \frac{m G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial^2}{(\partial t)^2} T(t)}{24\, |G(x)|^{2m} G(x)^2\, T(t)} + \frac{m G(x)^{2m} P(x)^{-n} T(t)^{2m} T(t)^{-4m} \frac{\partial}{\partial x} G(x)^2}{24\, G(x)^3\, H(x)^2} \\
& - \frac{m G(x)^{2m} P(x)^{-n} T(t)^{2m} T(t)^{-4m} \frac{\partial^2}{(\partial x)^2} G(x)}{24\, G(x)^2\, H(x)^2} + \frac{m G(x)^{2m} P(x)^{-n} T(t)^{2m} T(t)^{-4m} \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial x} H(x)}{24\, G(x)^2\, H(x)^3} - \frac{n G(x)^{2m} P(x)^{-n} T(t)^{2m} T(t)^{-4m} \frac{\partial}{\partial x} H(x) \frac{\partial}{\partial x} P(x)}{48\, G(x)\, H(x)^3\, P(x)} \\
& - \frac{n G(x)^{2m} P(x)^{-n} T(t)^{2m} T(t)^{-4m} \frac{\partial}{\partial x} P(x)^2}{48\, G(x)\, H(x)^2\, P(x)^2} + \frac{n G(x)^{2m} P(x)^{-n} T(t)^{2m} T(t)^{-4m} \frac{\partial^2}{(\partial x)^2} P(x)}{48\, G(x)\, H(x)^2\, P(x)} - \frac{G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial t} T(t)^2}{96\, |G(x)|^{2m} G(x)^2\, T(t)^2} \\
& - \frac{G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial^2}{(\partial t)^2} T(t)}{12\, |G(x)|^{2m} G(x)^2\, T(t)} - \frac{G(x)^{2m} P(x)^{-n} T(t)^{2m} T(t)^{-4m} \frac{\partial}{\partial x} G(x)^2}{32\, G(x)^3\, H(x)^2} - \frac{m G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial}{\partial x} G(x)^2}{24\, |G(x)|^{2m} G(x)^4\, H(x)^2} \\
& - \frac{G(x)^{2m} P(x)^{-n} T(t)^{2m} T(t)^{-4m} \frac{\partial^2}{(\partial x)^2} G(x)}{16\, G(x)^2\, H(x)^2} + \frac{m G(x)^{4m} P(x)^{-n} T(t)^{-4m} |T(t)|^{2m} |G(x)| \frac{\partial^2}{(\partial x)^2} G(x)}{24\, |G(x)|^{2m} G(x)^3\, H(x)^2} + \ldots
\end{aligned}
$$

$$
\begin{aligned}
NP14 = & \frac{m G(x)^{4m} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{\frac{\partial}{\partial t} T(t)} \frac{\partial}{\partial t} T(t)}{8\, P(x)^n\, T(t)^2\, |G(x)|} - \frac{m G(x)^{4m} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \frac{\partial}{\partial t} T(t)^2}{4\, P(x)^n\, T(t)^2\, |G(x)|} \\
& + \frac{m G(x)^{4m} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{T(t)} \frac{\partial}{\partial t} T(t)^2}{8\, P(x)^n\, T(t)^3\, |G(x)|} + \frac{m G(x)^{4m} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{\frac{\partial}{\partial t} T(t)} \frac{\partial}{\partial x} G(x)}{8\, P(x)^n\, G(x)\, H(x)\, T(t)\, |G(x)|} \\
& - \frac{m G(x)^{4m} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{\frac{\partial}{\partial x} G(x)} \frac{\partial}{\partial t} T(t)}{8\, P(x)^n\, G(x)\, H(x)\, T(t)\, |G(x)|} - \frac{m G(x)^{4m} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{G(x)} \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial t} T(t)}{8\, P(x)^n\, G(x)^2\, H(x)\, T(t)\, |G(x)|} \\
& + \frac{m G(x)^{4m} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{T(t)} \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial t} T(t)}{8\, P(x)^n\, G(x)\, H(x)\, T(t)^2\, |G(x)|} - \frac{m G(x)^{4m} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{\frac{\partial}{\partial x} G(x)} \frac{\partial}{\partial x} G(x)}{8\, P(x)^n\, G(x)^2\, H(x)^2\, |G(x)|} \\
& + \frac{m G(x)^{4m} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \frac{\partial}{\partial x} G(x)^2}{4\, P(x)^n\, G(x)^2\, H(x)^2\, |G(x)|} - \frac{m G(x)^{4m} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{G(x)} \frac{\partial}{\partial x} G(x)^2}{8\, P(x)^n\, G(x)^3\, H(x)^2\, |G(x)|} \\
& + \frac{G(x)^{4m} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \overline{\frac{\partial}{\partial x} G(x)} \frac{\partial}{\partial t} T(t)}{16\, P(x)^n\, G(x)\, H(x)\, T(t)\, |G(x)|} - \frac{G(x)^{4m} T(t)^{2m} T(t)^{-6m} |G(x)|^{2m} |G(x)|^{-4m} |T(t)|^{2m} \frac{\partial}{\partial x} G(x) \frac{\partial}{\partial t} T(t)}{8\, P(x)^n\, G(x)\, H(x)\, T(t)\, |G(x)|} + \ldots
\end{aligned}
$$

$$
\begin{aligned}
NP16 = & -\frac{mnG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\overline{\frac{\partial}{\partial t}T(t)}\frac{\partial}{\partial t}T(t)}{8T(t)^{2}|G(x)|} \\
& + \frac{mnG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\frac{\partial}{\partial t}T(t)^{2}}{4T(t)^{2}|G(x)|} - \frac{mnG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\overline{T(t)}\frac{\partial}{\partial t}T(t)^{2}}{8T(t)^{3}|G(x)|} \\
& + \frac{mnG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\overline{\frac{\partial}{\partial x}G(x)}\frac{\partial}{\partial t}T(t)}{8G(x)H(x)T(t)|G(x)|} \\
& - \frac{mnG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\frac{\partial}{\partial x}G(x)\frac{\partial}{\partial t}T(t)}{4G(x)H(x)T(t)|G(x)|} \\
& + \frac{mnG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\overline{G(x)}\frac{\partial}{\partial x}G(x)\frac{\partial}{\partial t}T(t)}{8G(x)^{2}H(x)T(t)|G(x)|} \\
& - \frac{nG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\frac{\partial^{2}}{(\partial t)^{2}}T(t)}{4T(t)|G(x)|} + \frac{mG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\overline{\frac{\partial}{\partial t}T(t)}\frac{\partial}{\partial x}P(x)}{8H(x)P(x)T(t)|G(x)|} \\
& - \frac{nG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\overline{\frac{\partial}{\partial x}G(x)}\frac{\partial}{\partial t}T(t)}{16G(x)H(x)T(t)|G(x)|} \\
& + \frac{nG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\frac{\partial}{\partial x}G(x)\frac{\partial}{\partial t}T(t)}{8G(x)H(x)T(t)|G(x)|} \\
& - \frac{nG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\overline{G(x)}\frac{\partial}{\partial x}G(x)\frac{\partial}{\partial t}T(t)}{16G(x)^{2}H(x)T(t)|G(x)|} \\
& - \frac{nG(x)^{4m}P(x)^{-n}T(t)^{2m}T(t)^{-6m}|G(x)|^{2m}|G(x)|^{-4m}|T(t)|^{2m}\frac{\partial}{\partial x}H(x)\frac{\partial}{\partial t}T(t)}{8H(x)^{2}T(t)|G(x)|} - \dots
\end{aligned}
$$

$$
\begin{aligned}
NP17 = {} & \frac{2(3mT(t)^{2m+1}|T(t)|^{2m}\overline{\partial_t T(t)}\,\partial_t T(t) + (3mT(t)^{2m}\overline{T(t)} + (n^2-10m-1)T(t)^{2m+1})|T(t)|^{2m}(\partial_t T(t))^2)G(x)^{4m+3}|G(x)|^{2m}H(x)^3P(x)^2}{48\,G(x)^3H(x)^3\,T(t)^{6m+3}|G(x)|^{4m+1}P(x)^{n+2}} \\
& + \frac{8(m+1)T(t)^{2m+2}|T(t)|^{2m}\partial_t^2 T(t)\,G(x)^{4m+3}|G(x)|^{2m}H(x)^3P(x)^2}{48\,G(x)^3H(x)^3\,T(t)^{6m+3}|G(x)|^{4m+1}P(x)^{n+2}} \\
& - \frac{2((n^2-4m-1)T(t)^{4m+1}(\partial_t T(t))^2 + 4(m+1)T(t)^{4m+2}\partial_t^2 T(t))G(x)^{2m+2}|G(x)|^{4m+1}H(x)^3P(x)^2}{48\,G(x)^3H(x)^3\,T(t)^{6m+3}|G(x)|^{4m+1}P(x)^{n+2}} \\
& - \frac{3(2m-1)G(x)^{4m+1}T(t)^{2m+2}|T(t)|^{2m}H(x)^2P(x)^2\overline{G(x)}\,\partial_t T(t)\,|G(x)|^{2m}\partial_x G(x)}{48\,G(x)^3H(x)^3\,T(t)^{6m+3}|G(x)|^{4m+1}P(x)^{n+2}} \\
& - \frac{6(mT(t)^{2m+2}|T(t)|^{2m}\overline{\partial_t T(t)} + (mT(t)^{2m+1}\overline{T(t)} - (4m-1)T(t)^{2m+2})|T(t)|^{2m}\partial_t T(t))G(x)^{4m+2}|G(x)|^{2m}H(x)^2P(x)^2\partial_x G(x)}{48\,G(x)^3H(x)^3\,T(t)^{6m+3}|G(x)|^{4m+1}P(x)^{n+2}} \\
& + \frac{4n(G(x)^{4m+3}T(t)^{2m+3}|G(x)|^{2m}|T(t)|^{2m}P(x) - G(x)^{2m+2}T(t)^{4m+3}|G(x)|^{4m+1}P(x))\partial_x H(x)\,\partial_x P(x)}{48\,G(x)^3H(x)^3\,T(t)^{6m+3}|G(x)|^{4m+1}P(x)^{n+2}} \\
& + \frac{2(2n-1)(G(x)^{4m+3}T(t)^{2m+3}|G(x)|^{2m}|T(t)|^{2m}H(x) - G(x)^{2m+2}T(t)^{4m+3}|G(x)|^{4m+1}H(x))(\partial_x P(x))^2}{48\,G(x)^3H(x)^3\,T(t)^{6m+3}|G(x)|^{4m+1}P(x)^{n+2}} \\
& - \frac{3(2m-1)(G(x)^{4m+2}T(t)^{2m+2}|G(x)|^{2m}|T(t)|^{2m}H(x)^2P(x)^2\partial_t T(t) - G(x)^{4m+1}T(t)^{2m+3}|G(x)|^{2m}|T(t)|^{2m}H(x)P(x)^2\partial_x G(x))\overline{\partial_x G(x)}}{48\,G(x)^3H(x)^3\,T(t)^{6m+3}|G(x)|^{4m+1}P(x)^{n+2}} \\
& + \frac{4(2m-3)(G(x)^{4m+2}T(t)^{2m+3}|G(x)|^{2m}|T(t)|^{2m}H(x)P(x)^2 - G(x)^{2m+1}T(t)^{4m+3}|G(x)|^{4m+1}H(x)P(x)^2)\partial_x^2 G(x)}{48\,G(x)^3H(x)^3\,T(t)^{6m+3}|G(x)|^{4m+1}P(x)^{n+2}} \\
& - \frac{2(3G(x)^{6m+2}|T(t)|^{4m}H(x)P(x)^2T(t)^2|G(x)|\partial_t T(t) - 3G(x)^{4m+3}T(t)^{2m+2}|G(x)|^{2m}|T(t)|^{2m}H(x)P(x)^2\partial_t T(t))\partial_x H(x)}{48\,G(x)^3H(x)^3\,T(t)^{6m+3}|G(x)|^{4m+1}P(x)^{n+2}} \\
& + \frac{2(3G(x)^{6m+1}|T(t)|^{4m}P(x)^2T(t)^3|G(x)| - (4m-3)G(x)^{4m+2}T(t)^{2m+3}|G(x)|^{2m}|T(t)|^{2m}P(x)^2 + 2(2m-3)G(x)^{2m+1}T(t)^{4m+3}|G(x)|^{4m+1}P(x)^2)\partial_x G(x)\,\partial_x H(x)}{48\,G(x)^3H(x)^3\,T(t)^{6m+3}|G(x)|^{4m+1}P(x)^{n+2}} \\
& - \frac{4n(G(x)^{4m+3}T(t)^{2m+3}|G(x)|^{2m}|T(t)|^{2m}H(x)P(x) - G(x)^{2m+2}T(t)^{4m+3}|G(x)|^{4m+1}H(x)P(x))\partial_x^2 P(x)}{48\,G(x)^3H(x)^3\,T(t)^{6m+3}|G(x)|^{4m+1}P(x)^{n+2}}
\end{aligned}
$$

$NP18 = 0$

Bianchi Identities

$BI1 = 0$

$$
\begin{aligned}
BI2 = & \frac{(16m^2nT(t)^{2m}\overline{T(t)} + (n^3 - (32m^2+4m+1)n)T(t)^{2m+1})|T(t)|^{3m}(\partial_t T(t))^3\, G(x)^{6m+3}|G(x)|^{2m}H(x)^5P(x)^3}{64\sqrt{P(x)^n}\, G(x)^4H(x)^5\, T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{n+3}/\sqrt{2}} \\
& - \frac{4(2mnT(t)^{2m+1}\overline{T(t)} - (9m+1)nT(t)^{2m+2})|T(t)|^{3m}\partial_t T(t)\,\partial_t^2 T(t)\, G(x)^{6m+3}|G(x)|^{2m}H(x)^5P(x)^3 + 8nT(t)^{2m+3}|T(t)|^{3m}\partial_t^3 T(t)\, G(x)^{6m+3}|G(x)|^{2m}H(x)^5P(x)^3}{64\sqrt{P(x)^n}\, G(x)^4H(x)^5\, T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{n+3}/\sqrt{2}} \\
& + \frac{8(2m^2nT(t)^{2m+1}|T(t)|^{3m}(\partial_t T(t))^2 - mnT(t)^{2m+2}|T(t)|^{3m}\partial_t^2 T(t))\overline{\partial_t T(t)}\, G(x)^{6m+3}|G(x)|^{2m}H(x)^5P(x)^3}{64\sqrt{P(x)^n}\, G(x)^4H(x)^5\, T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{n+3}/\sqrt{2}} \\
& - \frac{((n^3-(4m+1)n)T(t)^{4m+1}|T(t)|^m(\partial_t T(t))^3 + 4(5m+1)nT(t)^{4m+2}|T(t)|^m\partial_t T(t)\,\partial_t^2 T(t) - 8nT(t)^{4m+3}|T(t)|^m\partial_t^3 T(t))G(x)^{4m+2}|G(x)|^{4m+1}H(x)^5P(x)^3}{64\sqrt{P(x)^n}\, G(x)^4H(x)^5\, T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{n+3}/\sqrt{2}} \\
& - \frac{5(2n+3)(G(x)^{6m+3}T(t)^{2m+4}|G(x)|^{2m}|T(t)|^{3m}H(x)^2 - G(x)^{4m+2}T(t)^{4m+4}|G(x)|^{4m+1}|T(t)|^m H(x)^2)(\partial_x P(x))^3}{64\sqrt{P(x)^n}\, G(x)^4H(x)^5\, T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{n+3}/\sqrt{2}} \\
& - \frac{8(G(x)^{6m+3}T(t)^{2m+4}|G(x)|^{2m}|T(t)|^{3m}H(x)^2P(x)^2 - G(x)^{4m+2}T(t)^{4m+4}|G(x)|^{4m+1}|T(t)|^m H(x)^2P(x)^2)\partial_x^3 P(x)}{64\sqrt{P(x)^n}\, G(x)^4H(x)^5\, T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{n+3}/\sqrt{2}} + \cdots
\end{aligned}
$$

$$
\begin{aligned}
BI3 = & -\frac{(16m^2nT(t)^{2m}\overline{T(t)} + (n^3 - (32m^2+4m+1)n)T(t)^{2m+1})|T(t)|^{3m}(\partial_t T(t))^3\, G(x)^{6m+3}|G(x)|^{2m}H(x)^5P(x)^3}{64\sqrt{P(x)^n}\, G(x)^4H(x)^5\, T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{n+3}/\sqrt{2}} \\
& + \frac{4(2mnT(t)^{2m+1}\overline{T(t)} - (9m+1)nT(t)^{2m+2})|T(t)|^{3m}\partial_t T(t)\,\partial_t^2 T(t)\, G(x)^{6m+3}|G(x)|^{2m}H(x)^5P(x)^3 + 8nT(t)^{2m+3}|T(t)|^{3m}\partial_t^3 T(t)\, G(x)^{6m+3}|G(x)|^{2m}H(x)^5P(x)^3}{64\sqrt{P(x)^n}\, G(x)^4H(x)^5\, T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{n+3}/\sqrt{2}} \\
& - \frac{8(2m^2nT(t)^{2m+1}|T(t)|^{3m}(\partial_t T(t))^2 - mnT(t)^{2m+2}|T(t)|^{3m}\partial_t^2 T(t))\overline{\partial_t T(t)}\, G(x)^{6m+3}|G(x)|^{2m}H(x)^5P(x)^3}{64\sqrt{P(x)^n}\, G(x)^4H(x)^5\, T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{n+3}/\sqrt{2}} \\
& + \frac{((n^3-(4m+1)n)T(t)^{4m+1}|T(t)|^m(\partial_t T(t))^3 + 4(5m+1)nT(t)^{4m+2}|T(t)|^m\partial_t T(t)\,\partial_t^2 T(t) - 8nT(t)^{4m+3}|T(t)|^m\partial_t^3 T(t))G(x)^{4m+2}|G(x)|^{4m+1}H(x)^5P(x)^3}{64\sqrt{P(x)^n}\, G(x)^4H(x)^5\, T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{n+3}/\sqrt{2}} \\
& - \frac{5(2n+3)(G(x)^{6m+3}T(t)^{2m+4}|G(x)|^{2m}|T(t)|^{3m}H(x)^2 - G(x)^{4m+2}T(t)^{4m+4}|G(x)|^{4m+1}|T(t)|^m H(x)^2)(\partial_x P(x))^3}{64\sqrt{P(x)^n}\, G(x)^4H(x)^5\, T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{n+3}/\sqrt{2}} \\
& + \frac{8(G(x)^{6m+3}T(t)^{2m+4}|G(x)|^{2m}|T(t)|^{3m}H(x)^2P(x)^2 - G(x)^{4m+2}T(t)^{4m+4}|G(x)|^{4m+1}|T(t)|^m H(x)^2P(x)^2)\partial_x^3 P(x)}{64\sqrt{P(x)^n}\, G(x)^4H(x)^5\, T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{n+3}/\sqrt{2}} + \cdots
\end{aligned}
$$

$$BI4 = 0$$

$$
\begin{aligned}
BI5 = {} & \frac{(4(mn^2-16m^2-m)T(t)^{2m}\overline{T(t)}-((16m+11)n^2-160m^2-60m-11)T(t)^{2m+1})|T(t)|^{3m}(\partial_t T(t))^3\,G(x)^{6m+4}|G(x)|^{2m}H(x)^5P(x)^3}{192\sqrt{P(x)^n}\,G(x)^5H(x)^5\,T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{n+3}/\sqrt{2}} \\
& + \frac{4(4(m^2+m)T(t)^{2m+1}\overline{T(t)}-(16m^2-5n^2+31m+6)T(t)^{2m+2})|T(t)|^{3m}\partial_t T(t)\,\partial_t^2 T(t)\,G(x)^{6m+4}|G(x)|^{2m}H(x)^5P(x)^3}{192\sqrt{P(x)^n}\,G(x)^5H(x)^5\,T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{n+3}/\sqrt{2}} \\
& + \frac{4(2n^2+3n-5)(G(x)^{6m+4}T(t)^{2m+4}|G(x)|^{2m}|T(t)|^{3m}H(x)^2-G(x)^{4m+3}T(t)^{4m+4}|G(x)|^{4m+1}|T(t)|^{m}H(x)^2)(\partial_x P(x))^3}{192\sqrt{P(x)^n}\,G(x)^5H(x)^5\,T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{n+3}/\sqrt{2}} \\
& + \frac{8(2m-3)(G(x)^{6m+3}T(t)^{2m+4}|G(x)|^{2m}|T(t)|^{3m}H(x)P(x)^3-G(x)^{4m+2}T(t)^{4m+4}|G(x)|^{4m+1}|T(t)|^{m}H(x)P(x)^3)\partial_x G(x)\,\partial_x^2 H(x)}{192\sqrt{P(x)^n}\,G(x)^5H(x)^5\,T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{n+3}/\sqrt{2}} + \cdots
\end{aligned}
$$

$$
\begin{aligned}
BI6 = {} & -\frac{(4(\sqrt{2}mn^2-16\sqrt{2}m^2-\sqrt{2}m)T(t)^{2m}\overline{T(t)}-((16\sqrt{2}m+11\sqrt{2})n^2-160\sqrt{2}m^2-60\sqrt{2}m-11\sqrt{2})T(t)^{2m+1})|T(t)|^{3m}(\partial_t T(t))^3\,G(x)^{6m+4}|G(x)|^{2m}H(x)^5P(x)^3}{192\sqrt{P(x)^n}\,G(x)^5H(x)^5\,T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{n+3}} \\
& - \frac{4((\sqrt{2}mn^2-16\sqrt{2}m^2-\sqrt{2}m)T(t)^{2m+1}|T(t)|^{3m}(\partial_t T(t))^2+4(\sqrt{2}m^2+\sqrt{2}m)T(t)^{2m+2}|T(t)|^{3m}\partial_t^2 T(t))\overline{\partial_t T(t)}\,G(x)^{6m+4}|G(x)|^{2m}H(x)^5P(x)^3}{192\sqrt{P(x)^n}\,G(x)^5H(x)^5\,T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{n+3}} \\
& + \frac{4(2\sqrt{2}n^2+3\sqrt{2}n-5\sqrt{2})(G(x)^{6m+4}T(t)^{2m+4}|G(x)|^{2m}|T(t)|^{3m}H(x)^2-G(x)^{4m+3}T(t)^{4m+4}|G(x)|^{4m+1}|T(t)|^{m}H(x)^2)(\partial_x P(x))^3}{192\sqrt{P(x)^n}\,G(x)^5H(x)^5\,T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{n+3}} \\
& + \frac{8(2\sqrt{2}m-3\sqrt{2})(G(x)^{6m+3}T(t)^{2m+4}|G(x)|^{2m}|T(t)|^{3m}H(x)P(x)^3-G(x)^{4m+2}T(t)^{4m+4}|G(x)|^{4m+1}|T(t)|^{m}H(x)P(x)^3)\partial_x G(x)\,\partial_x^2 H(x)}{192\sqrt{P(x)^n}\,G(x)^5H(x)^5\,T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{n+3}} + \cdots
\end{aligned}
$$

$$BI7 = 0$$

$$BI8 = 0$$

$$
\begin{aligned}
BI9 = {} & -\frac{2(5(\sqrt{2}mn^2-4\sqrt{2}m^2-\sqrt{2}m)T(t)^{2m}\overline{T(t)}-4((3\sqrt{2}m+\sqrt{2})n^2-12\sqrt{2}m^2-7\sqrt{2}m-\sqrt{2})T(t)^{2m+1})|T(t)|^{3m}(\partial_t T(t))^3\,G(x)^{6m+4}|G(x)|^{2m}H(x)^5P(x)^3}{128\sqrt{P(x)^n}\,G(x)^5H(x)^5\,T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{n+3}} \\
& - \frac{2(5(\sqrt{2}mn^2-4\sqrt{2}m^2-\sqrt{2}m)T(t)^{2m+1}|T(t)|^{3m}(\partial_t T(t))^2+4(\sqrt{2}m^2+2\sqrt{2}m)T(t)^{2m+2}|T(t)|^{3m}\partial_t^2 T(t))\overline{\partial_t T(t)}\,G(x)^{6m+4}|G(x)|^{2m}H(x)^5P(x)^3}{128\sqrt{P(x)^n}\,G(x)^5H(x)^5\,T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{n+3}} \\
& - \frac{2(2\sqrt{2}n^2+3\sqrt{2}n-10\sqrt{2})(G(x)^{6m+4}T(t)^{2m+4}|G(x)|^{2m}|T(t)|^{3m}H(x)^2-G(x)^{4m+3}T(t)^{4m+4}|G(x)|^{4m+1}|T(t)|^{m}H(x)^2)(\partial_x P(x))^3}{128\sqrt{P(x)^n}\,G(x)^5H(x)^5\,T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{n+3}} \\
& + \frac{2(2(4\sqrt{2}m^2+\sqrt{2}m-3\sqrt{2})G(x)^{4m+1}T(t)^{4m+3}|G(x)|^{4m+1}|T(t)|^{m}H(x)^3P(x)^3\partial_t T(t)-(3(4\sqrt{2}m^2-3\sqrt{2}m)T(t)^{2m+3}|T(t)|^{3m}\overline{\partial_t T(t)}}{128\sqrt{P(x)^n}\,G(x)^5H(x)^5\,T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{n+3}} \\
& - \frac{4(2\sqrt{2}m-5\sqrt{2})(G(x)^{6m+3}T(t)^{2m+4}|G(x)|^{2m}|T(t)|^{3m}H(x)P(x)^3-G(x)^{4m+2}T(t)^{4m+4}|G(x)|^{4m+1}|T(t)|^{m}H(x)P(x)^3)\partial_x G(x)\,\partial_x^2 H(x)}{128\sqrt{P(x)^n}\,G(x)^5H(x)^5\,T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{n+3}} + \cdots
\end{aligned}
$$

$BI10 = 0$

$$BI11 = -\frac{2(5(\sqrt{2}mn^2 - 4\sqrt{2}m^2 - \sqrt{2}m)T(t)^{2m}\overline{T(t)} - 4((3\sqrt{2}m + \sqrt{2})n^2 - 12\sqrt{2}m^2 - 7\sqrt{2}m - \sqrt{2})T(t)^{2m+1})|T(t)|^{3m}(\partial_t T(t))^3\, G(x)^{6m+4}|G(x)|^{2m}H(x)^5 P(x)^3\, \sqrt{P(x)^n}}{128\, G(x)^5 H(x)^5\, T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{2n+3}}$$
$$- \frac{2(5(\sqrt{2}mn^2 - 4\sqrt{2}m^2 - \sqrt{2}m)T(t)^{2m+1}|T(t)|^{3m}(\partial_t T(t))^2 + 4(\sqrt{2}m^2 + 2\sqrt{2}m)T(t)^{2m+2}|T(t)|^{3m}\partial_t^2 T(t))\overline{\partial_t T(t)}\, G(x)^{6m+4}|G(x)|^{2m}H(x)^5 P(x)^3\, \sqrt{P(x)^n}}{128\, G(x)^5 H(x)^5\, T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{2n+3}}$$
$$+ \frac{2(2\sqrt{2}n^2 + 3\sqrt{2}n - 10\sqrt{2})(G(x)^{6m+4}T(t)^{2m+4}|G(x)|^{2m}|T(t)|^{3m}H(x)^2 - G(x)^{4m+3}T(t)^{4m+4}|G(x)|^{4m+1}|T(t)|^m H(x)^2)(\partial_x P(x))^3\, \sqrt{P(x)^n}}{128\, G(x)^5 H(x)^5\, T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{2n+3}}$$
$$+ \frac{4(2\sqrt{2}m - 5\sqrt{2})(G(x)^{6m+3}T(t)^{2m+4}|G(x)|^{2m}|T(t)|^{3m}H(x)P(x)^3 - G(x)^{4m+2}T(t)^{4m+4}|G(x)|^{4m+1}|T(t)|^m H(x)P(x)^3)\partial_x G(x)\, \partial_x^2 H(x)\, \sqrt{P(x)^n}}{128\, G(x)^5 H(x)^5\, T(t)^{8m+4}|G(x)|^{5m+1/2}P(x)^{2n+3}} + \cdots$$

Petrov Type

Type "I"